

Univariate Channel Fusion for Multivariate Time Series Classification

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Motivation

- Physiological signals (EEG, ECG, ECoG) and wearable sensors produce high-frequency multivariate time series
- These applications often require fast inference and deployment on resource-constrained devices
- Existing simple strategies (e.g., channel concatenation or channel-wise ensembles) either increase the input dimensionality or require multiple classifiers

Research question

Can a simple and aggregated univariate representation provide competitive accuracy while reducing computational cost?

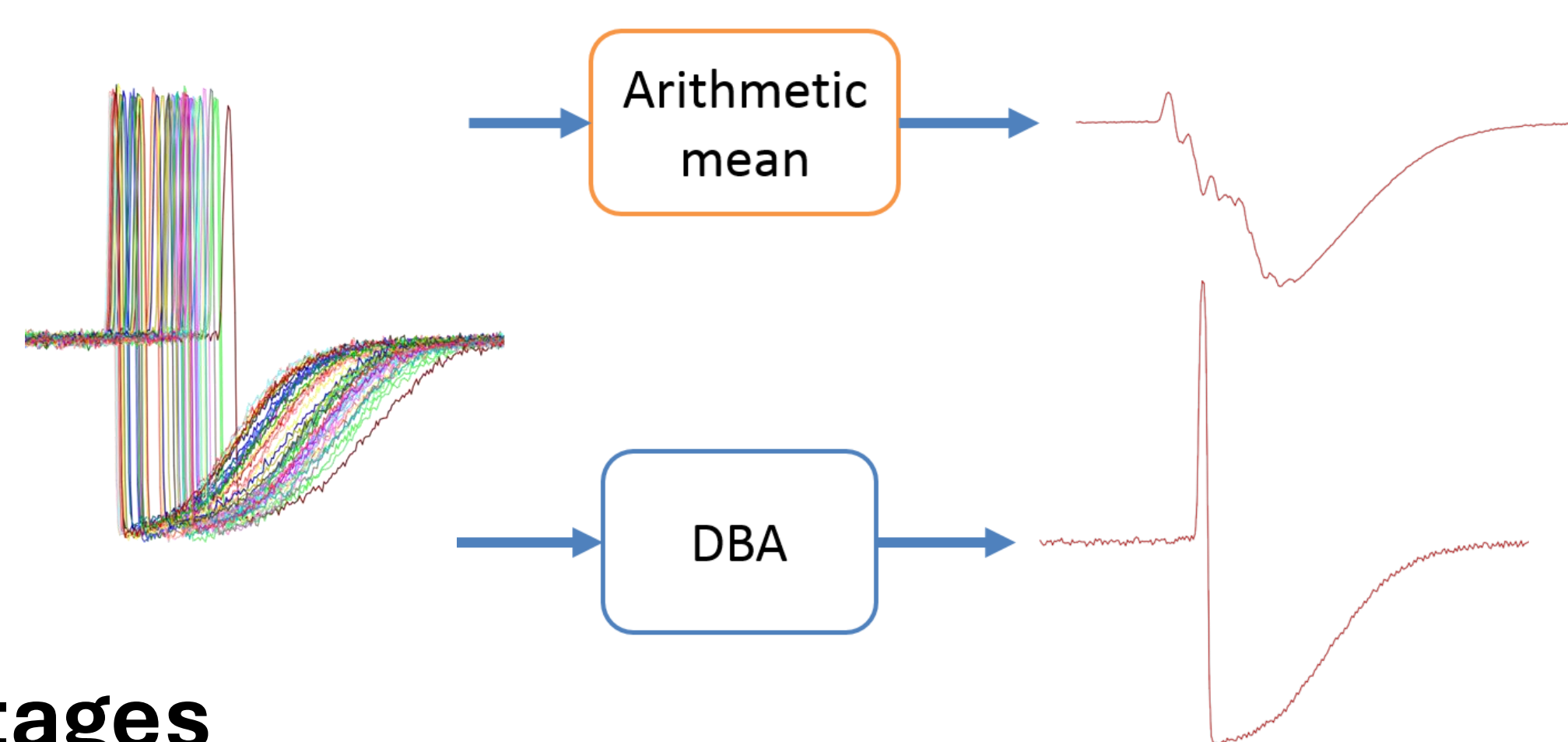
Univariate Channel Fusion (UCF)

A multivariate time series $X = \{x_1, \dots, x_n\}$, where each observation x_t contains the values of all channels, is converted into a univariate series by applying an aggregation function:

$$z_t = g(x_t)$$

Aggregation functions

- Mean** – arithmetic mean across channels
- Median** – median across channels
- DBA barycenter** – DTW-based series that preserves temporal alignments across channels



Advantages

- Compatible with any univariate model. For example, QUANT, an interval-based model
- Faster training and inference

Conclusions

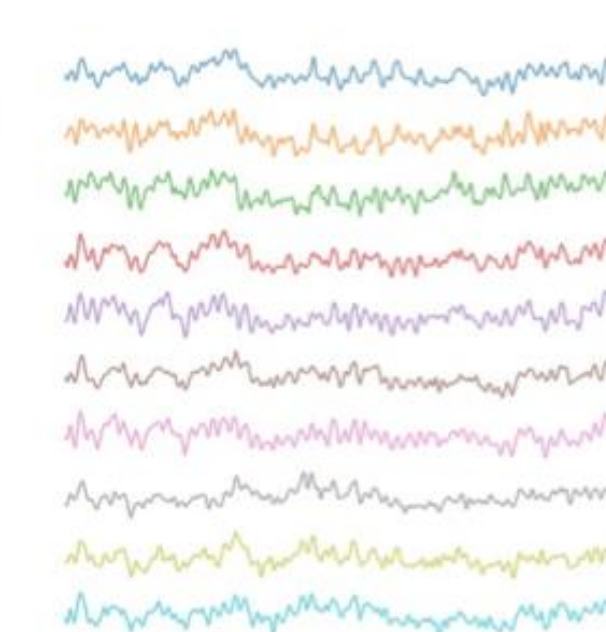
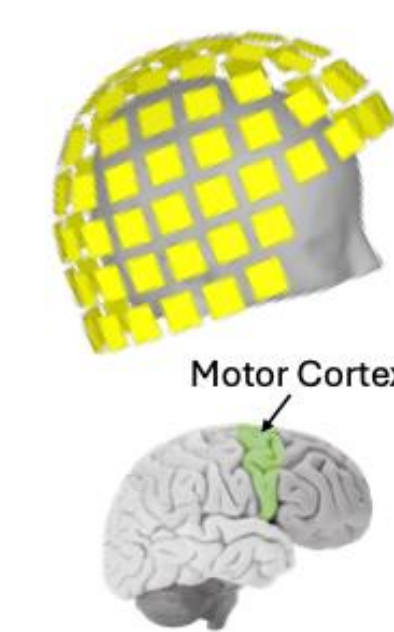
- UCF is a simple/effective strategy that enable the use of any univariate classifier. We suggest QUANT
- Achieves competitive or superior accuracy while substantially reducing computational cost
- Median fusion provides the best trade-off between accuracy and efficiency
- UCF is most suitable for datasets with highly correlated channels (same sensing modality); heterogeneous channels may lead to information loss (as showed in the additional experiments)
- Future work:** combining only groups of highly correlated channels

Experiments and case studies

- Cursor control via EEG
- Motion artifact effects in ECG
- Ethanol concentration via spectroscopy
- Motor imagery classification from ECoG
- Hand movement direction from MEG

Rival methods:

- WEASEL+MUSE
- TapNet
- QUANT
- DTW_D, DTW_I



Magnetoecephalography (MEG) signals

Results

Cursor control via EEG signals

Algorithm	Accuracy	Time
WEASEL+MUSE	79.18	3.32 min
TapNet	79.18	22.86 min
QUANT (concatenation)	80.54	9.27 s
QUANT (ensemble)	79.86	8.49 s
DTW _D	77.47	30.64 h
DTW _I	76.45	53.37 min
UCF (mean)	83.95	1.41 s
UCF (median)	87.37	1.53 s
UCF (barycenter)	87.71	2.98 min

Motion effects in ECG signals

Algorithm	Accuracy	Time
WEASEL+MUSE	40.00	5.70 s
TapNet	40.00	5.41 min
QUANT (concatenation)	46.66	0.87 s
QUANT (ensemble)	40.00	1.29 s
DTW _D	26.66	33.05 min
DTW _I	40.00	39.18 s
UCF (mean)	46.66	0.32 s
UCF (median)	60.00	0.32 s
UCF (barycenter)	66.66	11.68 s

Ethanol concentration detection

Algorithm	Accuracy	Time
WEASEL+MUSE	53.23	1.9 min
TapNet	27.37	54.9 min
QUANT (concatenation)	62.35	14.9 s
QUANT (ensemble)	65.77	13.2 s
DTW _D	34.22	100.4 h
DTW _I	30.79	1.5 h
UCF (mean)	68.44	4.3 s
UCF (median)	63.11	4.5 s
UCF (barycenter)	68.06	2.6 min

Hand movement direction classification

Algorithm	Accuracy	Time
WEASEL+MUSE	25.67	33.7 s
TapNet	33.78	6.4 min
QUANT (concatenation)	31.08	6.5 s
QUANT (ensemble)	33.78	8.4 s
DTW _D	22.97	56.1 min
DTW _I	31.08	2.7 min
UCF (mean)	35.13	0.8 s
UCF (median)	41.89	0.8 s
UCF (barycenter)	36.48	23.1 s

Motor imagery classification

Algorithm	Accuracy	Time
WEASEL+MUSE	48	1.9 h
TapNet	47	2.1 h
QUANT (concatenation)	51	23.2 min
QUANT (ensemble)	50	10.9 min
DTW _D	49	207.3 h
DTW _I	47	39.0 h
UCF (mean)	41	7.4 s
UCF (median)	52	7.8 s
UCF (barycenter)	43	1.7 h

The results show that **UCF achieves superior accuracy** compared to traditional baselines (DTW_D and DTW_I), deep learning models (TapNet), and specialized MTSC algorithms (WEASEL+MUSE), while maintaining **remarkable computational efficiency**

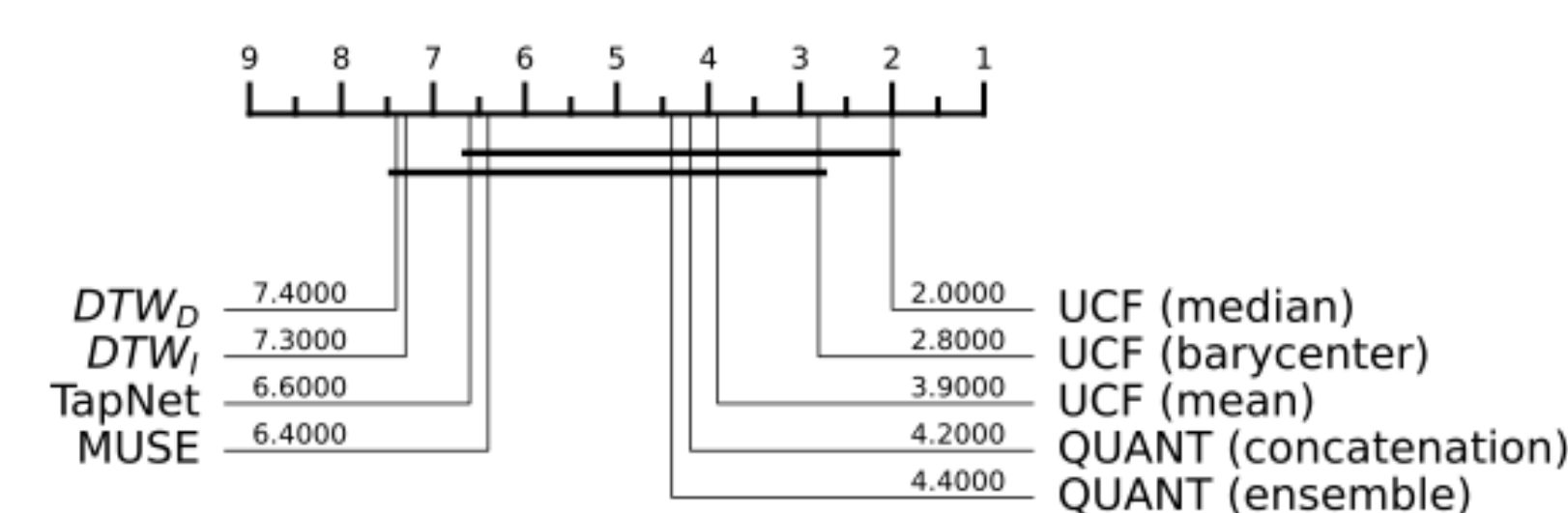
Statistical analysis

Accuracy

Critical difference diagrams

Nemenyi test ($\alpha = 0.1$)

5 datasets (case studies)

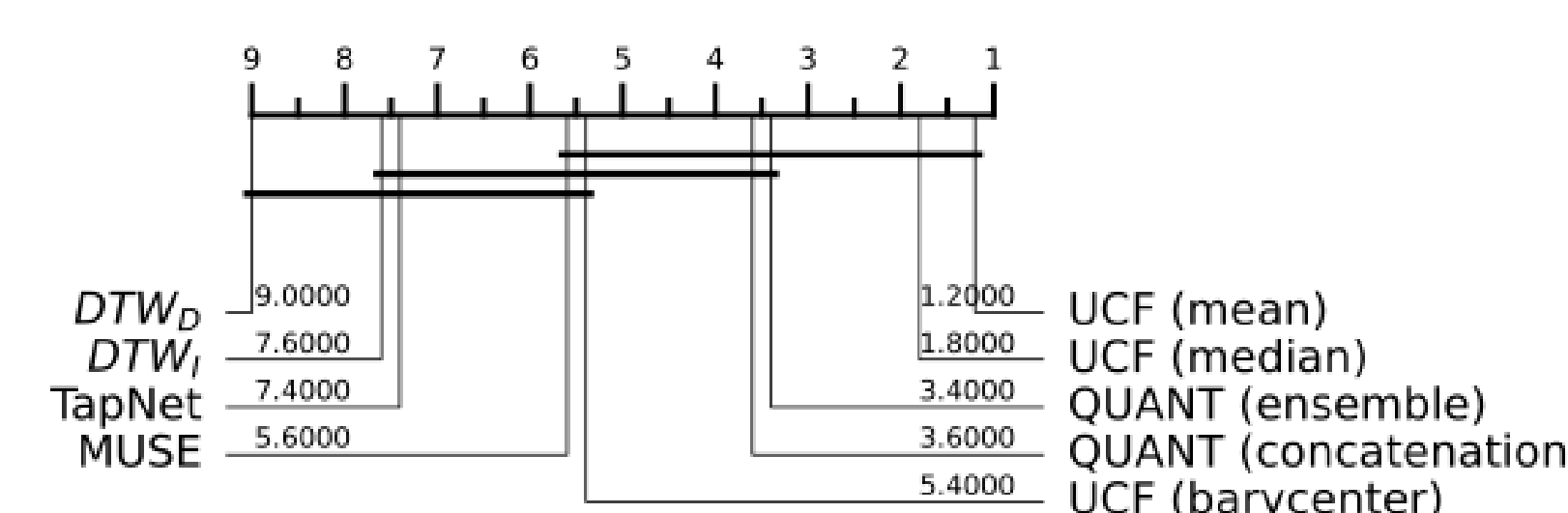


Time (5 runs)

AMD Ryzen 9 7950X

16-core CPU

28 GB of RAM



Supporting website
(codes, data, results)

